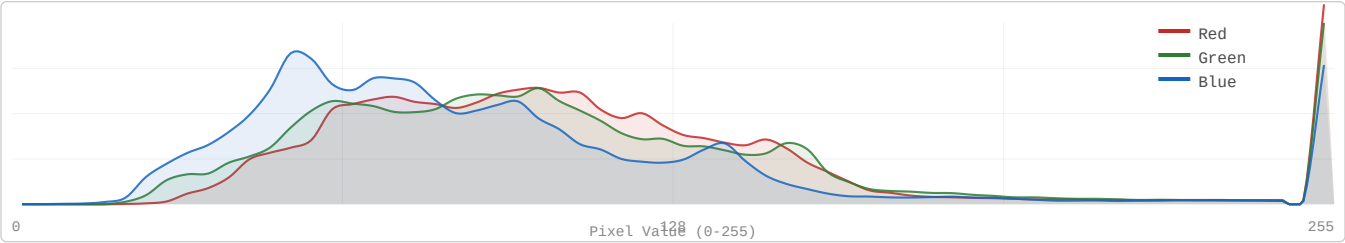
Decorrelation Gap: KODIM24

768x512 | 24-bit RGB | PNG (lossless)

White House in Snow | Near-degenerate (PC1 = 97.36%)

1. RGB Channel Distribution



2. Inter-Channel Correlation: RGB vs YCbCr

RGB (Original)

	R	G	B
R	1.0000	0.9761	0.9389
G	0.9761	1.0000	0.9649
B	0.9389	0.9649	1.0000

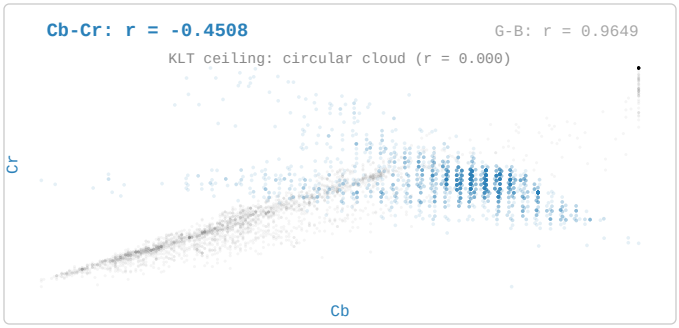
Cb-Cr: -0.451

(KLT ceiling = 0.000)

YCbCr (BT.601)

	Y	Cb	Cr
Y	1.0000	-0.1577	-0.1048
Cb	-0.1577	1.0000	-0.4508
Cr	-0.1048	-0.4508	1.0000

3. Cb vs Cr Scatter with G-B background overlay



4. Pairwise Decorrelation Gap

Channel Pair	RGB	YCbCr (BT.601)	Residual |r|
R-G → Y-Cb	0.976092	-0.157658	0.1577
R-B → Y-Cr	0.938867	-0.104759	0.1048
G-B → Cb-Cr	0.964925	-0.450835	0.4508
Avg |r|	0.959961	0.237750	0.2377

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5. Decorrelation Efficiency & Blue Independence

Decorrelation Efficiency

75.2% High

0% BT.601 leaves all — 100% KLT ceiling

Blue Channel Independence (Baetzel 2026a)

3.8%

0% Fully predictable — 100% Fully independent

6. 4:2:0 Chroma Subsampling — Information Cost

Channel	PSNR (dB)
R	42.14 dB
G	46.88 dB
B	36.75 dB
Overall	40.10 dB

Method: BT.601 RGB→YCbCr | 2x box downsample Cb,Cr | nearest upsample | BT.601 inverse

7. PCA Baseline Reference (Baetzel 2026a)

Eigendecomposition

Eigenvector Loadings

	R	G	B	Eigenvalue
PC1	-0.5728	-0.5911	-0.5678	8413.50
PC2	0.6349	0.1181	-0.7635	175.85
PC3	-0.5184	0.7979	-0.3077	52.42

8. PCA Derived Metrics

Condition Number

160.51

PC1/PC2: 47.84 | PC2/PC3: 3.35

Blue Channel Independence

3.8%

Variance not captured by PC1

PC1 Dominant Channel

G

Loading: G (0.59)

9. Redundancy Profile

Classification: Near-degenerate | CN: 160.51 | PC1: 97.4%

RGB Correlation: Avg |r| = 0.959961 | R-G: 0.9761 | R-B: 0.9389 | G-B: 0.9649

YCbCr Residual: Avg |r| = 0.237750 | Y-Cb: -0.1577 | Y-Cr: -0.1048 | Cb-Cr: -0.4508

Efficiency: 75.2% of KLT ceiling | Gap: 0.2377 avg |r| remaining

Blue Channel: 3.8% independent | 4:2:0 PSNR: 40.10 dB

References

[1] Baetzel, J. (2026a). "Statistical Characterization of Inter-Channel Redundancy Structure"
[2] Wallace, G.K. "The JPEG still picture compression standard." IEEE Trans. CE 38(1), 1992.
[3] ITU-R Recommendation BT.601: Studio Encoding Parameters, 1982.
[4] Watanabe, S. "Karhunen-Loeve Expansion and Factor Analysis," pp. 635-660, 1965.